

IEM KOLKATA
NUMBER SYSTEM

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Q1) The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that number?

- (a) 3 (b) 4 (c) 9 (d) Cannot be determined

Q2) A two-digit number is such that the product of the digits is 8. When 18 is added to the number, then the digits are reversed. The number is:

- (a) 18 (b) 24 (c) 42 (d) 81

Q3) $2^{25} + 2^{26} + 2^{27}$ is divisible by

- (a) 6 (b) 7 (c) 5 (d) 9

Q4) $32^5 + 2^{27}$ is divisible by

- (a) 3 (b) 7 (c) 10 (d) 11

Q5) Find the smallest number which when added to 4519 makes it exactly divisible by 30.

- (a) 6 (b) 11 (c) 10 (d) 13

Q6) Find the smallest number which on adding 19 to it is exactly divisible by 28, 36 and 45.

- (a) 1260 (b) 1241 (c) 1620 (d) 1214

Q7) How many numbers are there between 1 & 900 which are not divisible by 2, 3 or 5?

- (a) 240 (b) 245 (c) 250 (d) None

Q8) How many numbers upto 8700 are divisible by both 3 & 7?

- (a) 39 (b) 41 (c) 42 (d) 46

Q9) When 2^{256} is divided by 17, the remainder would be:

- (a) 2 (b) 0 (c) 5 (d) 1

Q10) What is the remainder when $(17^{25} + 19^{25})$ is divided by 18?

- (a) 0 (b) 1 (c) 3 (d) 9